Circles

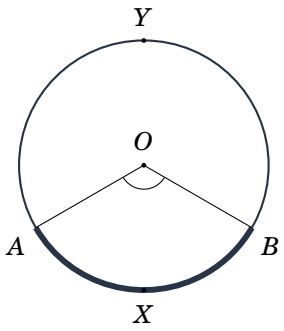
Class 3

Note for the teacher.

- **Aims.** Students name the minor and major arcs cut off by a chord and the angle each subtends at the centre. They discover, then explain, that the angle at the centre is double the angle at a point on the remaining part of the circle. They use this to show that angles in the same segment are equal and that the angle in a semicircle is 90°.
- **NCERT links.** Theorem 9 and its corollary keep their NCERT numbers. The questions include NCERT Exercise Set 5.6 and End-of-Chapter Questions 2, 6, 20 and 24.
- **Prior work.** Isosceles triangles (base angles equal), the angle sum of a triangle and the exterior angle property. Class 2: circumcircles.
- **Materials.** Two A4 sheets and the geometry box (protractor, compass, set square) for each student.
- **Method.** The *Solve first* problems P2 to P4 compute the same relation three times, in the three positions of the centre. Let students state the pattern before Theorem 9 is shown.

1 Arcs

Two points A and B on a circle cut it into two **arcs**. The smaller one is the **minor arc** and the larger one is the **major arc**. If AB is a diameter, the two arcs are equal; each is a **semicircle**.



The angle that an arc **subtends at the centre** is the angle swept as the radius turns from OA to OB along that arc. The minor arc AXB subtends $\angle AOB$. The major arc AYB subtends the reflex angle AOB, which is $360° - \angle AOB$.

Solve first.

P1. In a circle with centre O, $\angle AOB = 100°$.

(a) Which arc subtends 100° at the centre: the minor arc or the major arc?

(b) What angle does the other arc subtend at the centre?

(c) What angle does a semicircle subtend at the centre?

Answers.

P1. (a) The minor arc. (b) The major arc subtends the reflex angle $360° - 100° = 260°$. (c) $180°$: a straight angle.

2 The angle at the centre and the angle on the circle

The angle that arc AB **subtends at a point** C on the rest of the circle is $\angle ACB$. In P2 to P4, O is the centre and CD is a diameter. For each problem, draw a rough figure: the circle, its centre O, the diameter CD, the point A (and B), and the segments CA and OA (and CB and OB). Mark the given angles, and mark every angle you find.

Solve first.

P2. A is a point on the circle, and $\angle OCA = 25°$.

(a) Why is $\angle OAC = 25°$?
(b) $\angle AOD$ is an exterior angle of $\triangle OAC$. Find it.
(c) Compare $\angle AOD$ with $\angle ACD$.

P3. A and B lie on opposite sides of the diameter CD, with $\angle OCA = 20°$ and $\angle OCB = 30°$.

(a) Use P2 to find $\angle AOD$ and $\angle BOD$.
(b) Find $\angle AOB$ and $\angle ACB$.
(c) Compare $\angle AOB$ with $\angle ACB$.

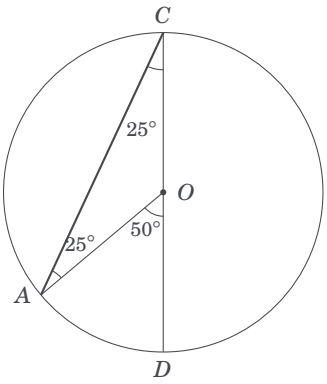
P4. A and B lie on the same side of the diameter CD, with $\angle OCA = 40°$ and $\angle OCB = 15°$.

(a) Find $\angle AOD$ and $\angle BOD$.
(b) Find $\angle AOB$ and $\angle ACB$.
(c) Compare $\angle AOB$ with $\angle ACB$. What pattern do P2, P3 and P4 show?

Answers.

P2.

(a) $OA = OC$ (radii), so the angles opposite them are equal.
(b) $\angle AOD = \angle OAC + \angle OCA = 25° + 25° = 50°$ (exterior angle).
(c) $\angle AOD = 50° = 2 \times 25° = 2\angle ACD$.

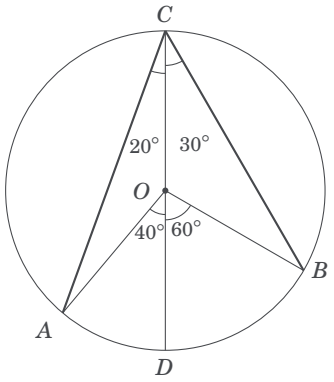


P3.

(a) As in P2: $\angle AOD = 2 \times 20^\circ = 40^\circ$ and $\angle BOD = 2 \times 30^\circ = 60^\circ$.

(b) $\angle AOB = 40^\circ + 60^\circ = 100^\circ$ and $\angle ACB = 20^\circ + 30^\circ = 50^\circ$.

(c) $\angle AOB = 2\angle ACB$.

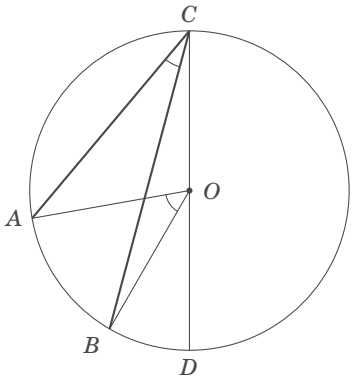


P4.

(a) As in P2: $\angle AOD = 2 \times 40^\circ = 80^\circ$ and $\angle BOD = 2 \times 15^\circ = 30^\circ$.

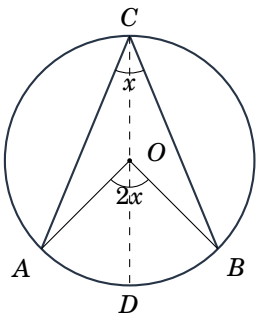
(b) $\angle AOB = 80^\circ - 30^\circ = 50^\circ$ and $\angle ACB = 40^\circ - 15^\circ = 25^\circ$.

(c) $\angle AOB = 2\angle ACB$. In all three problems the angle at the centre is double the angle at C.



Note for the teacher. P2 is the key step: P3 adds two copies of it and P4 subtracts one from the other. These are the three positions of the centre: on one arm of the angle, inside the angle and outside it. NCERT treats the second and third cases.

Theorem 9 (NCERT Theorem 9). *The angle subtended by an arc at the centre of the circle is double the angle subtended by the arc at any point on the circle outside the arc.*



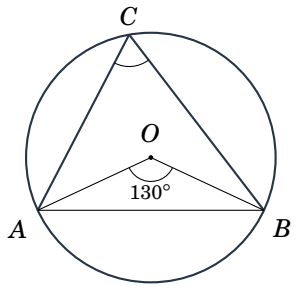
Why it works.

Given: arc AB of a circle with centre O; C is a point on the circle outside the arc.
To show: $\angle AOB = 2\angle ACB$.

1. Draw the diameter CD. $OA = OC$ (radii), so $\angle OAC = \angle OCA$. The exterior angle of $\triangle OAC$ is $\angle AOD = \angle OAC + \angle OCA = 2\angle OCA$.
2. In the same way, $\angle BOD = 2\angle OCB$.
3. If O lies inside $\angle ACB$, add: $\angle AOB = \angle AOD + \angle BOD = 2(\angle OCA + \angle OCB) = 2\angle ACB$.
4. If O lies outside $\angle ACB$, subtract: $\angle AOB = \angle AOD - \angle BOD = 2(\angle OCA - \angle OCB) = 2\angle ACB$.

The same steps work when the arc is a major arc; then $\angle AOB$ is a reflex angle.

Example 1. In a circle with centre O, $\angle AOB = 130°$, and C is a point on the major arc AB. Find $\angle ACB$ and $\angle OAB$.



Given: $\angle AOB = 130°$; C is on the major arc.
To find: $\angle ACB$ and $\angle OAB$.

1. C is outside the minor arc AB, which subtends $130°$ at O. So $\angle ACB = \dfrac{130°}{2} = 65°$ (Theorem 9).
2. $OA = OB$ (radii), so $\angle OAB = \angle OBA = \dfrac{180° - 130°}{2} = 25°$.

Answer: $\angle ACB = 65°$ and $\angle OAB = 25°$.

3 Two consequences

Solve first.

P5. In a circle with centre O, $\angle AOB = 80°$. C, D and E are three points on the major arc AB. Draw a rough figure: the circle, its centre O, the radii OA and OB, and the points C, D and E, each joined to A and to B.

(a) Find $\angle ACB$, $\angle ADB$ and $\angle AEB$.

(b) Does the answer depend on where C is on the major arc?

P6. AB is a diameter of a circle with centre O, and C is any other point on the circle.

(a) What angle does the semicircle not containing C subtend at O?

(b) Find $\angle ACB$.

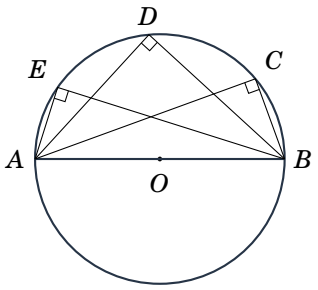
Answers.

P5. (a) Each is $\dfrac{80^\circ}{2} = 40^\circ$ (Theorem 9). (b) No. Every point of the major arc gives the same angle.

P6. (a) 180°. (b) $\angle ACB = \dfrac{180^\circ}{2} = 90^\circ$.

The part of the circle on one side of a chord, together with the chord, is a **segment**. P5 shows that *angles in the same segment are equal*: each is half the same angle at the centre.

Corollary of Theorem 9 (NCERT Corollary). *The angle subtended by a diameter at any point on the circle is* 90°.



Why it works.

Given: AB is a diameter of a circle with centre O; C is any other point on the circle.
To show: $\angle ACB = 90^\circ$.

1. The semicircle not containing C subtends a straight angle, 180°, at O.
2. By Theorem 9, $\angle ACB = \dfrac{180^\circ}{2} = 90^\circ$.

Things to Remember

The angle at the centre is double the angle on the circle, for the same arc.
Angles in the same segment are equal.
The angle in a semicircle is a right angle.

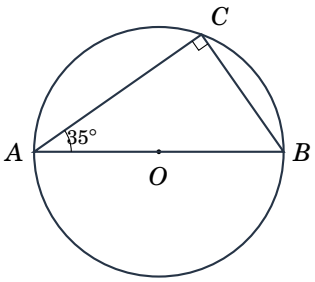
Solve first.

P7. On an A4 sheet, draw a circle of radius 4 cm and a chord AB of length 6 cm. On the larger side of AB, mark a point C on the circle, a point P inside the circle and a point Q outside the circle. Join CA and CB, PA and PB, and QA and QB. Measure $\angle ACB$, $\angle APB$ and $\angle AQB$ with a protractor. Which is the largest? Which is the smallest?

Answers.

P7. $\angle APB > \angle ACB > \angle AQB$: a point inside the circle gives a larger angle, and a point outside gives a smaller one. Only points on the arc give the angle $\angle ACB$. Class 4 uses this fact.

Example 2. AB is a diameter of a circle with centre O, and C is a point on the circle with $\angle CAB = 35°$. Find $\angle ABC$.



Given: AB is a diameter; $\angle CAB = 35°$.
To find: $\angle ABC$.

1. $\angle ACB = 90°$ (angle in a semicircle).
2. $\angle ABC = 180° - 90° - 35° = 55°$ (angle sum of a triangle).

Answer: $\angle ABC = 55°$.

For every question in Exercise 3A that has no figure, draw a rough figure: the circle, its centre O, and every point, chord and angle named in the question. Mark the given angles on it.

Exercise 3A

1. An arc of a circle subtends an angle of $70°$ at the centre. Find the angle it subtends at a point on the rest of the circle.

   Solution.

   Given: an arc subtends $70°$ at the centre.
   To find: the angle it subtends at a point on the rest of the circle.

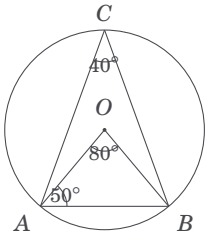
   1. Angle at the centre $= 70° = 2\times$ (angle at the point) (Theorem 9).
   2. So angle at the point $= \dfrac{70°}{2} = 35°$ (Theorem 9).

   Answer: $35°$.

2. In a circle with centre O, C is a point on the major arc AB and $\angle ACB = 40°$. Find $\angle AOB$ and $\angle OAB$.

   Solution.

   Given: C is a point on the major arc AB of a circle with centre O, and $\angle ACB = 40°$.
   To find: $\angle AOB$ and $\angle OAB$.

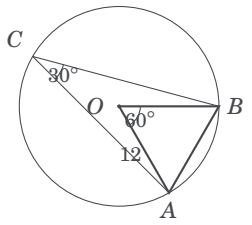
   

   1. $\angle AOB = 2\angle ACB = 2 \times 40° = 80°$ (Theorem 9).
   2. $OA = OB$ (radii), so $\angle OAB = \angle OBA$ (angles opposite equal sides).
   3. $\angle OAB = \dfrac{180° - 80°}{2} = 50°$ (angle sum of a triangle).

   Answer: $\angle AOB = 80°$ and $\angle OAB = 50°$.

3. In a circle with centre O and radius 12 cm, $\angle AOB = 60°$. Find the length of the chord AB, and the angle that AB subtends at a point on the major arc.

Solution.



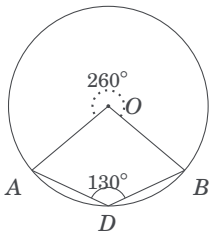
Given: a circle with centre O and radius 12 cm, and $\angle AOB = 60°$.
To find: the length of the chord AB and the angle it subtends at a point on the major arc.

1. $OA = OB = 12$ cm (radii), so $\angle OAB = \angle OBA$ (angles opposite equal sides).
2. $\angle OAB = \angle OBA = \dfrac{180° - 60°}{2} = 60°$ (angle sum of a triangle).
3. All three angles are 60°, so $\triangle OAB$ is equilateral and $AB = 12$ cm (equilateral triangle).
4. At a point on the major arc, AB subtends $\dfrac{60°}{2} = 30°$ (Theorem 9).

Answer: $AB = 12$ cm and the angle is 30°.

4. In a circle with centre O, $\angle AOB = 100°$, and D is a point on the *minor* arc AB. Ravi writes $\angle ADB = 50°$. Find his mistake, and find $\angle ADB$.

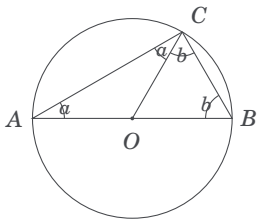
Solution.



Ravi is wrong. D lies *on* the minor arc, so it is outside the major arc, and the major arc subtends the reflex angle $360° - 100° = 260°$ at O. So $\angle ADB = \dfrac{260°}{2} = 130°$ (Theorem 9).
Answer: $\angle ADB = 130°$; Ravi halved the angle of the wrong arc.

5. AB is a diameter of a circle and C is any other point on the circle. Find $\angle ACB$ and explain your reasoning without using Theorem 9: join OC and use two isosceles triangles.

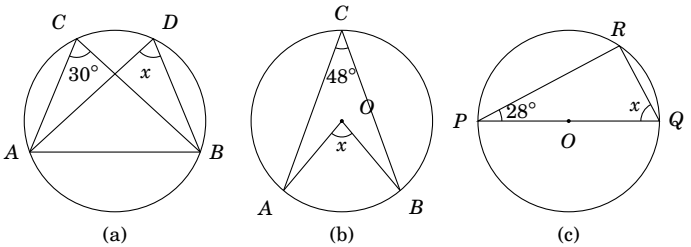
Solution.



Given: AB is a diameter of a circle with centre O, and C is any other point on the circle.
To prove: $\angle ACB = 90°$.
Construction: join OC.
Proof:

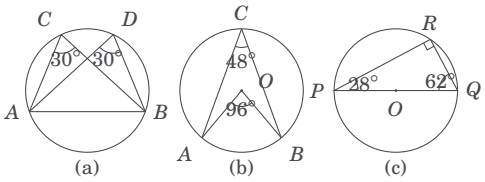
1. $OA = OC = OB$ (radii), so $\angle OAC = \angle OCA$ and $\angle OBC = \angle OCB$ (angles opposite equal sides).
2. Let $\angle OAC = \angle OCA = a$ and $\angle OBC = \angle OCB = b$.
3. In $\triangle ABC$, $a + (a + b) + b = 180°$ (angle sum of a triangle), so $2(a + b) = 180°$ and $a + b = 90°$.
4. $\angle ACB = a + b = 90°$ (from the angle sum of $\triangle ABC$).
5. Hence proved.

6. Find x in each figure. O is the centre.



Solution.

Given: O is the centre in each figure.
To find: x in each figure.



(a) $\angle ADB = \angle ACB$ (angles in the same segment), so $x = 30°$.

(b) $x = 2 \times 48° = 96°$ (Theorem 9).

(c) $\angle PRQ = 90°$ (angle in a semicircle), so $x = 180° - 90° - 28° = 62°$ (angle sum of a triangle).

Answer: (a) $x = 30°$, (b) $x = 96°$, (c) $x = 62°$.

7. A carpenter wants to check that a wooden piece has a true semicircular edge. She places the corner of a set square on the curved edge so that its two edges pass through the ends of the straight edge. She repeats this at several places. Explain the test.

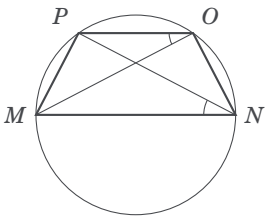
Solution.

The test is correct. The corner of the set square is a right angle, and the angle in a semicircle is 90° at *every* point of the curve, so the corner touches the curve at every position. If the corner leaves a gap at some place, the edge is not a true semicircle there.

8. $MNOP$ is a quadrilateral inscribed in a circle, and MN is a diameter. What can you say about $\angle MOP$ and $\angle MNP$? Explain.

Solution.

$\angle MOP = \angle MNP$. Both stand on the arc MP and lie in the same segment, so they are equal (angles in the same segment). Also $\angle MON = \angle MPN = 90°$ (angles in a semicircle), since MN is a diameter.



9. A and B are points on a circle.

   (a) Can there be points X and Y on the circle, on the same side of AB, with $\angle AXB \neq \angle AYB$?

   (b) If X and Y are on the circle and $\angle AXB = \angle AYB$, must X and Y be on the same side of AB?

   (c) If $\angle AXB = \angle AYB$, and X and Y are not on the circle, does the circle through A, B and X also pass through Y?

Solution.

(a) No. Points X and Y on the same side of AB lie on the same arc, so they are in the same segment and $\angle AXB = \angle AYB$ (angles in the same segment).

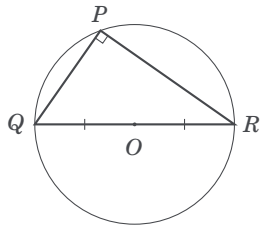
(b) No. If AB is a diameter, every point of the circle on either side gives 90° (angle in a semicircle).

(c) Yes, if X and Y are on the same side of AB: the circle through A, B and X also passes through Y. (This is Theorem 10, proved in Class 4.)

10. In Class 2 we saw by construction that the circumcentre of a right-angled triangle is the midpoint of its hypotenuse. Prove it. Let $\triangle PQR$ have $\angle P = 90^\circ$ and circumcentre O, and use Theorem 9 for the arc QR that does not contain P.

Solution.

Given: $\triangle PQR$ has $\angle P = 90^\circ$ and circumcentre O.
To prove: O is the midpoint of the hypotenuse QR.
Proof:



1. The arc QR not containing P subtends $\angle QPR = 90^\circ$ at P (given).
2. This arc subtends $2 \times 90^\circ = 180^\circ$ at the centre O (Theorem 9).
3. So $\angle QOR = 180^\circ$ (straight angle): Q, O and R lie on one straight line, and QR is a diameter.
4. $OQ = OR$ (radii), so O is the midpoint of the hypotenuse QR.
5. Hence proved.

11. A chord of a circle is equal to the radius. Find the angle it subtends at a point on the major arc.

Solution.

Given: a chord AB of a circle with centre O is equal to the radius.
To find: the angle it subtends at a point on the major arc.

1. $OA = OB$ (radii) and AB equals the radius (given), so $OA = OB = AB$ and $\triangle OAB$ is equilateral (equilateral triangle).
2. $\angle AOB = 60^\circ$ (angle of an equilateral triangle).
3. At a point on the major arc, AB subtends $\dfrac{60^\circ}{2} = 30^\circ$ (Theorem 9).

Answer: 30°.

Note for the teacher. Question 4 must be discussed in class: students apply "half" without checking which arc the point lies on. Question 9 prepares Class 4. If time is short, set Questions 8, 10 and 11 as homework.
